SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name : DAMASCUS

UFI	:	XGFE-VWF4-T00R-JWMX
Sales Number	:	6041554
Cust. Material	:	DAMASCUS

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Substance/Mixture	:	Fragrance for consumer product

1.3 Details of the supplier of the safety data sheet

Company	:	IFF (NEDERLAND) BV ZEVENHEUVELENWEG 60 5048 AN TILBURG
Telephone	:	+31134642211
Telefax	:	+31134636032
E-mail address of person responsible for the SDS	:	sds@iff.com

1.4 Emergency telephone number

ORFILA (INRS): + 33 (0)1 45 42 59 59

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Eye irritation, Category 2	H319: Causes serious eye irritation.
Skin sensitisation, Category 1	H317: May cause an allergic skin reaction.
Long-term (chronic) aquatic hazard, Category 2	H411: Toxic to aquatic life with long lasting effects.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

Hazard pictograms :

Signal word : Warning

Hazard statements :
H317 May cause an allergic skin reaction.
H319 Causes serious eye irritation.
H411 Toxic to aquatic life with long lasting effects.

Precautionary statements

Prevention:
P261 Avoid breathing mist or vapours.
P280 Wear protective gloves/ eye protection/ face protection.

Response:
P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention.
P337 + P313 If eye irritation persists: Get medical advice/ attention.
P391 Collect spillage.

Disposal:
P501 Dispose of contents/ container to an approved waste disposal plant.

Hazardous components which must be listed on the label:

63286-42-0	OXACYCLOHEPTADEC-10-EN-2-ONE
67874-81-1	OCTAHYDRO-METHOXY-TETRAMETHYL-METHANOAZULENE
120-57-0	HELIOTROPINE
127-51-5	Alpha-Isomethyl Ionone
5462-06-6	METHOXYHYDRATROPALDEHYDE
78-70-6	linalool
80-56-8, 7785-26-4	pin-2(3)-ene
5989-27-5	(R)-p-mentha-1,8-diene
1205-17-0	METHYLENEDIOXYPHENYL METHYLPROPANAL
13466-78-9	3,7,7-trimethylbicyclo[4.1.0]hept-3-ene
106-22-9, 7540-51-4	Citronellol

2.3 Other hazards

None reasonably foreseeable.

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.
Ecological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Toxicological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number		Concentration (% w/w)
tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans)	63500-71-0 405-040-6 603-101-00-3 01-0000015458-64, 01-2119455547-30	Eye Irrit. 2; H319	>= 20 - < 30
1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran	1222-05-5 214-946-9 603-212-00-7 01-2119488227-29	Aquatic Chronic 1; H410 Aquatic Acute 1; H400	>= 10 - < 20
(10E)-1-oxacycloheptadec-10-en-2-one	63286-42-0 814-308-5 01-2120768557-38	Skin Sens. 1B; H317 Aquatic Acute 1; H400	>= 2,5 - < 10
[3aR-(3aα,5aβ,9aα,9bβ)]-dodecahydro-3a,6,6,9a-tetramethylnaphtho[2,1-b]furan	6790-58-5 229-861-2 01-2120070784-50-	Aquatic Chronic 3; H412	>= 1 - < 2,5
[3R-(3α,3aβ,6a,7β,8aα)]-octahydro-6-methoxy-3,6,8,8-tetramethyl-1H-3a,7-methanoazulene	67874-81-1 267-510-5 01-2120228335-61	Skin Sens. 1B; H317 Aquatic Acute 1; H400 Aquatic Chronic 1; H410	>= 1 - < 2,5
piperonal	120-57-0 204-409-7 01-2119983608-21	Skin Sens. 1B; H317	>= 1 - < 10
RM of (3E)-3-methyl-4-(2,6,6-trimethylcyclohex-2-en-1-yl)but-3-en-2-one & (1E)-1-(2,6,6-trimethylcyclohex-2-en-1-yl)pent-1-en-3-one	127-51-5 204-846-3	Aquatic Chronic 2; H411 Skin Sens. 1B; H317 M-Factor (Acute aquatic toxicity): 1	>= 1 - < 2,5
1,4-dioxacyclohexadecane-5,16-dione	54982-83-1 259-423-6 01-2119524000-64	Aquatic Acute 1; H400 Aquatic Chronic 3; H412	>= 1 - < 2,5
vanillin	121-33-5 204-465-2 01-2119516040-60	Eye Irrit. 2; H319	>= 1 - < 10
3-(p-methoxyphenyl)-2-methylpropionaldehyde	5462-06-6 226-749-5 01-2120769416-45, 01-2120629103-67	Skin Sens. 1B; H317 M-Factor (Acute aquatic toxicity): 1	>= 1 - < 10

linalool	78-70-6 201-134-4 603-235-00-2 01-2119474016-42	Skin Irrit. 2; H315 Eye Irrit. 2; H319 Skin Sens. 1B; H317	>= 1 - < 10
methyl anthranilate	134-20-3 205-132-4	Eye Irrit. 2; H319	>= 1 - < 10
pin-2(3)-ene	80-56-8 201-291-9 01-2119519223-49	Flam. Liq. 3; H226 Asp. Tox. 1; H304 Skin Irrit. 2; H315 Skin Sens. 1B; H317 Aquatic Acute 1; H400 Aquatic Chronic 1; H410 Acute Tox. 4; H302 M-Factor (Acute aquatic toxicity): 1	>= 0,25 - < 1
(R)-p-mentha-1,8-diene	5989-27-5 227-813-5 601-029-00-7 01-2119529223-47	Skin Irrit. 2; H315 Skin Sens. 1B; H317 Aquatic Chronic 3; H412 Flam. Liq. 3; H226 Asp. Tox. 1; H304 Aquatic Acute 1; H400 M-Factor (Acute aquatic toxicity): 1	>= 0,25 - < 1
α-methyl-1,3-benzodioxole-5-propionaldehyde	1205-17-0 214-881-6	Skin Sens. 1B; H317 Aquatic Chronic 2; H411 Repr. 2; H361 M-Factor (Acute aquatic toxicity): 1	>= 0,25 - < 1
3,7,7-trimethylbicyclo[4.1.0]hept-3-ene	13466-78-9 236-719-3	Flam. Liq. 3; H226 Asp. Tox. 1; H304 Skin Sens. 1; H317 Skin Irrit. 2; H315 Aquatic Chronic 1; H410	>= 0,1 - < 0,25
citronellol	106-22-9 203-375-0 01-2119453995-23	Skin Irrit. 2; H315 Skin Sens. 1B; H317 Eye Irrit. 2; H319	>= 0,1 - < 1
cyclopentadecanone	502-72-7 207-951-2 01-2120766374-48	Aquatic Acute 1; H400 Aquatic Chronic 1; H410 M-Factor (Acute aquatic toxicity): 1 M-Factor (Chronic	>= 0,1 - < 0,25

		aquatic toxicity): 1	
indole	120-72-9 204-420-7	Acute Tox. 3; H311 Acute Tox. 4; H302 Eye Dam. 1; H318	>= 0,1 - < 1

For explanation of abbreviations see section 16.

SECTION 4: First aid measures

4.1 Description of first aid measures

General advice : Take Hazard and Precautionary phrases (section 2) into account.

If inhaled : Remove from exposure site to fresh air and keep at rest. If victim is unconscious, remove foreign bodies from the mouth. If victim has stopped breathing, give artificial respiration. Obtain medical advice.

In case of skin contact : Remove contaminated clothes. Wash thoroughly with water (and soap). Contact physician if symptoms persist.

In case of eye contact : Flush immediately with water for at least 15 minutes. Contact physician if symptoms persist.

If swallowed : Rinse mouth with water and obtain medical advice.

4.2 Most important symptoms and effects, both acute and delayed

Risks : May cause an allergic skin reaction.
Causes serious eye irritation.

4.3 Indication of any immediate medical attention and special treatment needed

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media : Carbondioxide, dry chemical, foam.

Unsuitable extinguishing media : Do not use a direct waterjet on burning material.

5.2 Special hazards arising from the substance or mixture

Specific hazards during firefighting : Water may be ineffective.

5.3 Advice for firefighters

Special protective equipment for firefighters : Wear self-contained breathing apparatus for firefighting if necessary.

Further information : Standard procedure for chemical fires.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions : Avoid inhalation and contact with skin and eyes. A self-contained breathing apparatus is recommended in case of a major spill.
Prevent spreading over a wide area (e.g. by containment or oil barriers).

6.2 Environmental precautions

Environmental precautions : Keep away from drains, surface- and groundwater and soil.

6.3 Methods and material for containment and cleaning up

Methods for cleaning up : Clean up spillage promptly. Remove ignition sources. Provide adequate ventilation. Avoid excessive inhalation of vapours. Gross spillages should be contained by use of sand or inert powder and disposed of according to the local regulations.

6.4 Reference to other sections

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advice on safe handling : Avoid excessive inhalation of concentrated vapors. Follow good manufacturing practices for housekeeping and personal hygiene. Wash any exposed skin immediately after any chemical contact, before breaks and meals, and at the end of each work period. Contaminated clothing and shoes should be thoroughly cleaned before re-use.

If appropriate, procedures used during the handling of this material should also be used when cleaning equipment or removing residual chemicals from tanks or other containers, especially when steam or hot water is used, as this may increase vapor concentrations in the workplace air. Where chemicals are openly handled, access should be restricted to properly trained employees.
Keep all heated processes at the lowest necessary temperature in order to minimize emissions of volatile chemicals into the air.

Advice on protection against fire and explosion : Keep away from ignition sources and naked flame.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : Store in a cool, dry, ventilated area away from heat sources. Keep containers upright and tightly closed when not in use.

7.3 Specific end use(s)

Specific use(s) : No information available.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Contains no substances with occupational exposure limit values.

Derived No Effect Level (DNEL) according to Regulation (EC) No. 1907/2006:

			Potential health effects	Value
tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans)	Workers	Inhalation	Long-term systemic effects	44,1 mg/m3
Remarks:	REACH data			
	Workers	Dermal	Long-term systemic effects	41,7 mg/kg bw/day
Remarks:	REACH data			
	Workers	Eye contact	Local effects	
Remarks:	Available hazard data do not enable the derivation of a DNEL for eye irritant effects.			
	General population	Inhalation	Long-term systemic effects	13 mg/m3
Remarks:	REACH data			
	General population	Dermal	Long-term systemic effects	25
Remarks:	REACH data			
	General population	Oral	Long-term systemic effects	7,5 mg/kg bw/day
Remarks:	REACH data			

Predicted No Effect Concentration (PNEC) according to Regulation (EC) No. 1907/2006:

Substance name	Environmental Compartment	Value
1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran	Fresh water	0,0044 mg/l
	Marine water	0,00044 mg/l
	Fresh water sediment	2 mg/kg dry weight (d.w.)
	Marine sediment	0,394 mg/kg dry weight (d.w.)
	Soil	0,31 mg/kg dry weight (d.w.)

8.2 Exposure controls

Engineering measures

Where appropriate, use closed systems to transfer and process this material.
If appropriate, isolate mixing rooms and other areas where this material is used or openly handled. Maintain these areas under negative air pressure relative to the rest of the plant.

Personal protective equipment

Eye protection : Use tight-fitting goggles, face shield or safety glasses with side shields if eye contact might occur.
Equipment should conform to EN 166

Hand protection

Material : Nitrile rubber
Break through time : > 60 min
Glove thickness : 0,38 mm

Material : Nitrile rubber
Break through time : > 10 min
Glove thickness : 0,1 mm

Remarks : Avoid skin contact. Use chemically resistant gloves. The selected protective gloves have to satisfy the specifications of Regulation (EU) 2016/425 and the standard EN 374 derived from it. Be aware that in daily use the durability of a chemical resistant protective glove can be notably shorter than the break through time measured according to EN 374, due to the numerous outside influences (e.g. temperature). Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion, and the contact time. The choice of an appropriate glove does not only depend on its material but also on other quality features and is different from one producer to the other.

Skin and body protection : Choose body protection in relation to its type, to the concentration and amount of dangerous substances, and to the specific work-place.
Personal protection through wearing a tightly closed chemical protection suit and a self-contained breathing apparatus.

Respiratory protection : Use local exhaust ventilation around open tanks and other open sources of potential exposures in order to avoid excessive inhalation, including places where this material is openly weighed or measured. In addition, use general dilution ventilation of the work area to eliminate or reduce possible worker exposures.
No respiratory protection is required during normal operations in a workplace where engineering controls such as adequate ventilation, etc. are sufficient.

If engineering controls and safe work practices are not sufficient, an approved, properly fitted respirator with organic vapor cartridges or canisters and particulate filters should be used:

a)while engineering controls and appropriate safe work prac-

		tices and/or procedures are being implemented; or b)during short term maintenance procedures when engineering controls are not in normal operation or are not sufficient; or c)if normal operational workplace vapor concentration in the air is increased due to heat ; d)during emergencies; or e)if engineering controls and operational practices are not sufficient to reduce airborne concentrations below an established occupational exposure limit.
Protective measures	:	To the extent deemed appropriate, implement pre-placement and regularly scheduled ascertainment of symptoms and spirometry testing of lung function for workers who are regularly exposed to this material. To the extent deemed appropriate, use an experienced air sampling expert to identify and measure volatile chemicals that could be present in the workplace air to determine potential exposures and to ensure the continuing effectiveness of engineering controls and operational practices to minimize exposure.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	:	liquid
Colour	:	pale greenish yellow to greenish yellow
Odour	:	conforms to standard
Odour Threshold	:	not determined
Melting point	:	not determined
Boiling point	:	not determined
Flammability	:	not determined
Upper explosion limit / Upper flammability limit	:	not determined
Lower explosion limit / Lower flammability limit	:	not determined
Flash point	:	127,00 °C Method: closed cup
Auto-ignition temperature	:	not determined
Decomposition temperature	:	not determined
pH	:	not determined
Viscosity, dynamic	:	not determined
Viscosity, kinematic	:	not determined
Water solubility	:	not determined
Solubility in other solvents	:	not determined
Partition coefficient: n-octanol/water	:	not determined

Vapour pressure : 0,03 hPa
Calculated

Relative density : 0,9875 - 0,9975

Density : not determined

9.2 Other information

No data available

SECTION 10: Stability and reactivity

10.1 Reactivity

No hazards to be specially mentioned.

10.2 Chemical stability

Stable under normal conditions.

10.3 Possibility of hazardous reactions

Hazardous reactions : Presents no significant reactivity hazard, by itself or in contact with water.

10.4 Conditions to avoid

Conditions to avoid : Direct sources of heat.

10.5 Incompatible materials

Materials to avoid : Avoid contact with strong acids, alkali or oxidizing agents.

10.6 Hazardous decomposition products

Carbon monoxide and unidentified organic compounds may be formed during combustion.

SECTION 11: Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Not classified based on available information.

Product:

Acute dermal toxicity : Acute toxicity estimate: > 2.000 mg/kg
Method: Calculation method

Skin corrosion/irritation

Not classified based on available information.

Components:

linalool:

Species : Rabbit
Exposure time : 4 h
Assessment : Causes skin irritation.
Method : OECD Test Guideline 404

Result	:	Skin irritation
GLP	:	yes
Test substance	:	(undiluted)
Remarks	:	REACH data

(10E)-1-oxacycloheptadec-10-en-2-one:

Species	:	human
Exposure time	:	48 h
Method	:	repeated insult patch test
Result	:	No skin irritation
Species	:	Rabbit
Exposure time	:	24 h
Result	:	No skin irritation

[3R-(3α,3aβ,6a,7β,8aα)]-octahydro-6-methoxy-3,6,8,8-tetramethyl-1H-3a,7-methanoazulene:

Species	:	reconstructed human epidermis (RhE)
Exposure time	:	1 h
Assessment	:	No skin irritation
Method	:	OECD Test Guideline 439
Result	:	No skin irritation
GLP	:	yes
Test substance	:	(undiluted)
Remarks	:	REACH data

Serious eye damage/eye irritation

Causes serious eye irritation.

<u>Components:</u>

tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans):

Species	:	Rabbit
Assessment	:	Causes serious eye irritation.
Method	:	Regulation (EC) No. 440/2008, Annex, B.5
Result	:	Eye irritation
GLP	:	yes
Remarks	:	REACH data

vanillin:

Remarks	:	No data available

linalool:

Species	:	Rabbit
Assessment	:	Causes serious eye irritation.
Method	:	OECD Test Guideline 405
Result	:	Eye irritation
GLP	:	no
Remarks	:	REACH data

Respiratory or skin sensitisation

Skin sensitisation

May cause an allergic skin reaction.

Respiratory sensitisation

Not classified based on available information.

Components:

linalool:

Test Type	:	Local lymph node assay (LLNA)
Species	:	Mouse
Assessment	:	The product is a skin sensitiser, sub-category 1B.
Method	:	OECD Test Guideline 429
Result	:	The product is a skin sensitiser, sub-category 1B.
GLP	:	yes
Remarks	:	REACH data

[3R-(3α,3aβ,6a,7β,8aα)]-octahydro-6-methoxy-3,6,8,8-tetramethyl-1H-3a,7-methanoazulene:

Test Type	:	Local lymph node assay (LLNA)
Species	:	Mouse
Assessment	:	The product is a skin sensitiser, sub-category 1B.
Method	:	OECD Test Guideline 429
Result	:	Causes sensitisation.
GLP	:	yes
Test substance	:	8.0% in petrolatum
Remarks	:	REACH data

Germ cell mutagenicity

Not classified based on available information.

Components:

tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans):

Genotoxicity in vitro	:	Test Type: reverse mutation assay Test system: TA1535 Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 471 Result: negative GLP: yes Remarks: REACH data Test Type: reverse mutation assay Test system: TA98 Metabolic activation: with and without metabolic activation Method: OECD Test Guideline 471 Result: negative GLP: yes Remarks: REACH data Test Type: reverse mutation assay

Test system: TA100
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: yes
Remarks: REACH data

Test Type: reverse mutation assay
Test system: TA1537
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: yes
Remarks: REACH data

Test Type: reverse mutation assay
Test system: WP2 uvrA
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative
GLP: yes
Remarks: REACH data

Genotoxicity in vivo : Test Type: In vivo micronucleus test
Species: Mouse (male and female)
Application Route: Oral
Method: Regulation (EC) No. 440/2008, Annex, B.12
Result: negative
GLP: yes
Remarks: REACH data

1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran:

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: Mutagenicity (Escherichia coli - reverse mutation assay)
Result: negative

Test Type: Chromosome aberration test in vitro
Metabolic activation: with and without metabolic activation
Method: OECD 473
Result: negative

Carcinogenicity

Not classified based on available information.

Reproductive toxicity

Not classified based on available information.

STOT - single exposure

Not classified based on available information.

STOT - repeated exposure

Not classified based on available information.

Repeated dose toxicity

Components:

1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran:

Species	:	Rat, male and female
NOAEL	:	>= 150 mg/kg
Application Route	:	Oral
Exposure time	:	90-day
Number of exposures	:	1x /day
Method	:	OECD 408

Aspiration toxicity

Not classified based on available information.

Components:

pin-2(3)-ene:

May be fatal if swallowed and enters airways.

11.2 Information on other hazards

Endocrine disrupting properties

Product:

Assessment	:	The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

SECTION 12: Ecological information

12.1 Toxicity

Components:

1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran:

Toxicity to fish	:	LC50 (Lepomis macrochirus (Bluegill sunfish)): 0,452 mg/l Exposure time: 21 d Test Type: flow-through test Method: OECD Test Guideline 204 GLP: yes
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 0,9 mg/l Exposure time: 48 h Test Type: semi-static test Method: OECD Test Guideline 202 GLP: yes

Toxicity to algae/aquatic plants	:	ErC50 (Pseudokirchneriella subcapitata (green algae)): > 0,854 mg/l Exposure time: 72 h Test Type: static test Method: OECD Test Guideline 201 EbC50 (Pseudokirchneriella subcapitata (green algae)): 0,723 mg/l Exposure time: 72 h Test Type: static test Method: OECD Test Guideline 201
Toxicity to fish (Chronic toxicity)	:	NOEC: 0,068 mg/l Exposure time: 36 d Species: Pimephales promelas (fathead minnow) Test Type: flow-through test Method: OECD 210 GLP: yes
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC: 0,111 mg/l Exposure time: 21 d Species: Daphnia magna (Water flea) Test Type: semi-static test Method: OECD 211

12.2 Persistence and degradability

<u>Components:</u>

tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans):

Biodegradability	:	Test Type: aerobic Inoculum: activated sludge, non-adapted Concentration: 30 mg/l Result: Inherently biodegradable. Biodegradation: 64,8 % Related to: Biochemical oxygen demand Exposure time: 60 d Method: OECD Test Guideline 301D GLP: no Remarks: REACH data

1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran:

Biodegradability	:	Result: Not readily biodegradable. Biodegradation: 2 % Exposure time: 28 d Method: Modified Sturm Test Test Type: aerobic Inoculum: activated sludge Concentration: 10 mg/l Result: Primary biodegradation Biodegradation: 33,8 % Exposure time: 28 d

Method: OECD 301 B
Remarks: see user defined free text

Test Type: aerobic
Inoculum: activated sludge, adapted
Concentration: 10,97 mg/l
Result: Primary biodegradation
Biodegradation: 28,3 %
Exposure time: 28 d
Method: OECD 301 B
Remarks: see user defined free text

(10E)-1-oxacycloheptadec-10-en-2-one:

Biodegradability : Biodegradation: 91,3 %
Exposure time: 28 d
Method: OECD Test Guideline 301F

12.3 Bioaccumulative potential

Components:

tetrahydro-2-isobutyl-4-methylpyran-4-ol, mixed isomers (cis and trans):

Partition coefficient: n-octanol/water : log Pow: 1,65 (23 °C)
Method: Regulation (EC) No. 440/2008, Annex, A.8
GLP: yes
Remarks: REACH data

1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran:

Bioaccumulation : Species: Lepomis macrochirus (Bluegill sunfish)
Exposure time: 28 d
Bioconcentration factor (BCF): 1.584
Method: OECD 305

Partition coefficient: n-octanol/water : log Pow: 5,3 (25 °C)
GLP: no
Remarks: REACH data

log Pow: 5,9 (25 °C)
Method: OECD Test Guideline 117
GLP: yes
Remarks: REACH data

12.4 Mobility in soil

Product:

Mobility : Remarks: No data available

Components:

1,3,4,6,7,8-hexahydro-4,6,6,7,8,8-hexamethylindeno[5,6-c]pyran:

Distribution among environ- : Koc: 24547, log Koc: 4,39 mental compartments

12.5 Results of PBT and vPvB assessment

Product:

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

12.6 Endocrine disrupting properties

Product:

Assessment : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

12.7 Other adverse effects

Product:

Additional ecological information : There is no data available for this product.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product : Dispose of according to local regulations. Avoid disposing into drainage systems and into the environment.

Contaminated packaging : Empty containers should be taken to an approved waste handling site for recycling or disposal.

SECTION 14: Transport information

14.1 UN number or ID number

ADR : UN 3082

IMDG : UN 3082

IATA : UN 3082

14.2 UN proper shipping name

ADR : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(HEXAMETHYLINDANOPYRAN)

IMDG : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.

		(HEXAMETHYLINDANOPYRAN)
IATA	:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (HEXAMETHYLINDANOPYRAN)

14.3 Transport hazard class(es)

ADR	:	9
IMDG	:	9
IATA	:	9

14.4 Packing group

ADR		
Packing group	:	III
Classification Code	:	M6
Hazard Identification Number	:	90
Labels	:	9
Tunnel restriction code	:	(-)
IMDG		
Packing group	:	III
Labels	:	9
EmS Code	:	F-A, S-F
IATA (Cargo)		
Packing instruction (cargo aircraft)	:	964
Packing instruction (LQ)	:	Y964
Packing group	:	III
Labels	:	Miscellaneous Dangerous Goods
IATA_P (Passenger)		
Packing instruction (passenger aircraft)	:	964
Packing instruction (LQ)	:	Y964
Packing group	:	III
Labels	:	Miscellaneous Dangerous Goods

14.5 Environmental hazards

ADR		
Environmentally hazardous	:	yes
IMDG		
Marine pollutant	:	yes (HEXAMETHYLINDANOPYRAN)
IATA (Passenger)		
Environmentally hazardous	:	yes
IATA (Cargo)		
Environmentally hazardous	:	yes

14.6 Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data

Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

14.7 Maritime transport in bulk according to IMO instruments

Not applicable for product as supplied.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

REACH - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles (Annex XVII)	:	Conditions of restriction for the following entries should be considered: Number on list 3 diisobutyl phthalate (Number on list 51d, 30)
REACH - Candidate List of Substances of Very High Concern for Authorisation (Article 59).	:	Not applicable
Regulation (EC) No 1005/2009 on substances that deplete the ozone layer	:	Not applicable
Regulation (EU) 2019/1021 on persistent organic pollutants (recast)	:	Not applicable
Regulation (EU) 2019/1148 on the marketing and use of explosives precursors	:	acetone
REACH - List of substances subject to authorisation (Annex XIV)	:	Not applicable

Regulation (EU) 2019/1148 on the marketing and use of explosives precursors

acetone (ANNEX II)

Reinforced medical supervision (R4624-18)	:	The product has no CMR properties

Other regulations:

Take note of Directive 94/33/EC on the protection of young people at work or stricter national regulations, where applicable.

15.2 Chemical safety assessment

A Chemical Safety Assessment is not required for this substance.

SECTION 16: Other information

Full text of H-Statements

H226	:	Flammable liquid and vapour.
H302	:	Harmful if swallowed.
H304	:	May be fatal if swallowed and enters airways.
H311	:	Toxic in contact with skin.
H315	:	Causes skin irritation.
H317	:	May cause an allergic skin reaction.
H318	:	Causes serious eye damage.
H319	:	Causes serious eye irritation.
H361	:	Suspected of damaging fertility or the unborn child.
H400	:	Very toxic to aquatic life.
H410	:	Very toxic to aquatic life with long lasting effects.
H411	:	Toxic to aquatic life with long lasting effects.
H412	:	Harmful to aquatic life with long lasting effects.

Full text of other abbreviations

Acute Tox.	:	Acute toxicity
Aquatic Acute	:	Short-term (acute) aquatic hazard
Aquatic Chronic	:	Long-term (chronic) aquatic hazard
Asp. Tox.	:	Aspiration hazard
Eye Dam.	:	Serious eye damage
Eye Irrit.	:	Eye irritation
Flam. Liq.	:	Flammable liquids
Repr.	:	Reproductive toxicity
Skin Irrit.	:	Skin irritation
Skin Sens.	:	Skin sensitisation

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of Very High Concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

Further information

Other information : In December 2003, the National Institute for Occupational Safety and Health ("NIOSH") published an Alert on preventing lung disease in workers who use or make flavorings [NIOSH Publication Number 2004-110].
In August 2004, the United States Flavor and Extract Manufacturers Association (FEMA) issued a report entitled "Respiratory Safety in the Flavor Manufacturing Workplace".
Both of these reports provide recommendations for reducing employee exposure and for medical surveillance in the workplace. The recommendations in these reports are generally applicable to the use of any chemical in the workplace and you are strongly urged to review both of these reports.
The report published by FEMA also contains a list of "high priority" chemicals. If any of these chemicals are present in this product at a concentration >= 1.0% due to an intentional addition by IFF, the chemical(s) will be identified in this safety data sheet.

According to Regulation (EC) No. 1907/2006 the information in this safety data sheet is based on the properties of the material known to IFF at the time the data sheet was issued. The safety data sheet is intended to provide information for a health and safety assessment of the material and the circumstances, under which it is packaged, stored or applied in the workplace. For such a safety assessment International Flavors & Fragrances holds no responsibility. This document is not intended for quality assurance purposes.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

FR / EN